



How to implement a DAS Listener in C#

Reference : RA 468

Welcome to the documentation for the **RA_0468 DAS Listener**.
This reference architecture demonstrates how to build a responsive Data Analytics Solution (DAS) that listens for messages from a test program and processes them dynamically.

General Technical Info

Component	Version
Language	C#
Framework	.NET 8
Environment	Windows 10
IGXL	> 10.30.10
MST	> 2016C
IDE	Visual Studio Code / Visual Studio 2022

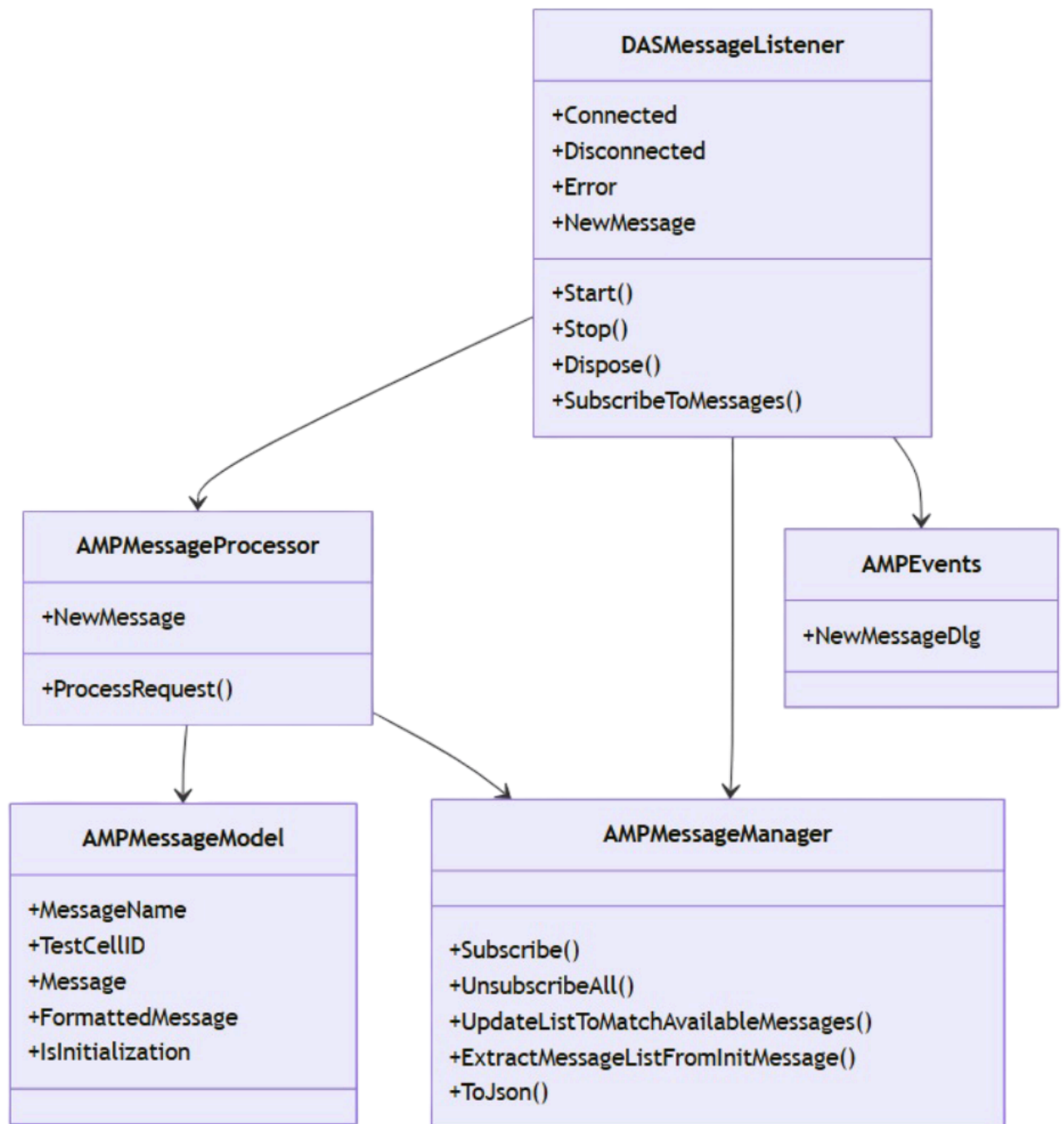
Key Features

- Real-time HTTP message reception using **HttpListener**
- Events for connection, disconnection, errors, and new message dispatch
- Structured message model including timestamps, URL segments, and raw/parsed payload
- Fully documented C# code using .NET 8

Architecture Overview

The DAS acts as a lightweight server that receives messages from a test cell. Each message is parsed, validated, and dispatched to an event system.

Project Structure Diagram



Example: Getting Started

Here's a quick example of how to instantiate and use the `DASMessageListener` class:

```

using Teradyne.Archimedes.AMPCommunication;
using Teradyne.Archimedes.DAS.Listener;

public class Program
{
    public static async Task Main(string[] args)
    {
        Console.WriteLine("Hello, Welcome to my DAS!");

        var listener = new DASMessageListener("http://localhost:3000/tems/");

        listener.Connected += () => Console.WriteLine("Connected to DAS.");
        listener.Disconnected += () => Console.WriteLine("Disconnected.");
        listener.Error += (ex) => Console.WriteLine($"Error: {ex.Message}");
        listener.NewMessage += (cellid, message) =>
        {
            Console.WriteLine($"New message from {cellid}: {message.MessageName}");
        };

        await listener.Start();
    }
}

```

This documentation is part of the Teradyne Archimedes Reference Architecture series.

Namespace Teradyne.Archimedes. AMPCommunication

Classes

[AMPEvents](#)

Provides utility definitions for the DAS system, including event delegate declarations.

Delegates

[AMPEvents.NewMessageDlg](#)

Delegate for handling new message events. Triggered when a new [AMPMessageModel](#) is received from a test cell.

Class AMPEvents

Namespace: [Teradyne.Archimedes.AMPCommunication](#)

Assembly: Teradyne.Archimedes.DAS.Listener.dll

Provides utility definitions for the DAS system, including event delegate declarations.

```
public class AMPEvents
```

Inheritance

[object](#)  ← AMPEvents

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  ,
[object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Delegate AMPEvents.NewMessageDlg

Namespace: [Teradyne.Archimedes.AMPCommunication](#)

Assembly: Teradyne.Archimedes.DAS.Listener.dll

Delegate for handling new message events. Triggered when a new [AMPMessageModel](#) is received from a test cell.

```
public delegate void AMPEvents.NewMessageDlg(string cellid, AMPMessageModel message)
```

Parameters

cellid [string](#)

The identifier of the test cell that received the message.

message [AMPMessageModel](#)

The AMP message payload.

Namespace Teradyne.Archimedes. AMPCommunication.Messages

Classes

[AMPMessageManager](#)

Manages AMP message subscriptions and interactions. It is mandatory to return a list of message names as a response to the INITIALIZATION message and recommended for each message.

Class AMPMessageManager

Namespace: [Teradyne.Archimedes.AMPCommunication.Messages](#)

Assembly: Teradyne.Archimedes.DAS.Listener.dll

Manages AMP message subscriptions and interactions. It is mandatory to return a list of message names as a response to the INITIALIZATION message and recommended for each message.

```
public class AMPMessageManager
```

Inheritance

[object](#) ← AMPMessageManager

Inherited Members

[object.Equals\(object\)](#), [object.Equals\(object, object\)](#), [object.GetHashCode\(\)](#), [object.GetType\(\)](#), [object.MemberwiseClone\(\)](#), [object.ReferenceEquals\(object, object\)](#), [object.ToString\(\)](#)

Properties

MessageList

Current list of subscribed messages.

```
public HashSet<string> MessageList { get; }
```

Property Value

[HashSet](#) <[string](#)>

Methods

ExtractMessageListFromInitMessage(AMPMessageModel)

Extracts the message list from an INITIALIZATION message.

```
public void ExtractMessageListFromInitMessage(AMPMessageModel message)
```


Parameters

message [AMPMessageModel](#)

The INITIALIZATION message model.

Subscribe(IEnumerable<string>)

Subscribes to multiple message names.

```
public void Subscribe(IEnumerable<string> messageNames)
```

Parameters

messageNames [IEnumerable](#) <[string](#)>

A collection of message names to subscribe to.

Subscribe(string)

Subscribes to a single message name if it is valid.

```
public bool Subscribe(string messageName)
```

Parameters

messageName [string](#)

The name of the message to subscribe to.

Returns

[bool](#)

True if subscribed, false if already present or invalid.

SubscribeAll()

Subscribes to all predefined AMP messages.

```
public void SubscribeAll()
```

ToJson()

Serializes the message list into a JSON string.

```
public string ToJson()
```

Returns

[string](#) 

A JSON string containing the MESSAGE_LIST.

UnsubscribeAll()

Clears the current message subscription list.

```
public void UnsubscribeAll()
```

UpdateListToMatchAvailableMessages(AMPMessageManager)

Intersects the message list with another manager's list. Only shared messages are kept.

```
public void UpdateListToMatchAvailableMessages(AMPMessageManager available)
```

Parameters

available [AMPMessageManager](#)

Manager with the available message list.

Namespace Teradyne.Archimedes. AMPCommunication.Models

Classes

[AMPMessageModel](#)

Represents an AMP message with metadata, test cell ID, timestamps, and JSON message formatting.

Class AMPMessageModel

Namespace: [Teradyne.Archimedes.AMPCommunication.Models](#)

Assembly: Teradyne.Archimedes.DAS.Listener.dll

Represents an AMP message with metadata, test cell ID, timestamps, and JSON message formatting.

```
public class AMPMessageModel
```

Inheritance

[object](#)  ← AMPMessageModel

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Properties

FormattedMessage

The JSON-formatted version of the message. Falls back to raw message if invalid.

```
public string FormattedMessage { get; }
```

Property Value

[string](#) 

IsInitialization

Determines whether the message name indicates an initialization command.

```
public bool IsInitialization { get; }
```

Property Value

[bool](#)

Message

The raw JSON message string. When set, attempts to format it with indentation.

```
public string Message { get; set; }
```

Property Value

[string](#)

MessageName

The name of the message extracted from the URL.

```
public string MessageName { get; set; }
```

Property Value

[string](#)

TestCellID

The identifier of the test cell associated with the message.

```
public string TestCellID { get; set; }
```

Property Value

[string](#)

TimeStamp

The timestamp of the message, either parsed or defaulting to UTC now.

```
public DateTime TimeStamp { get; set; }
```

Property Value

[DateTime](#) 

URL

The URL segments associated with the message.

```
public string[] URL { get; set; }
```

Property Value

[string](#)  []

Namespace Teradyne.Archimedes.DAS.Listener

Classes

[DASMessageListener](#)

DASMessageListener is responsible for starting and stopping the DAS HTTP listener, receiving incoming HTTP requests, and dispatching them to a processor. It manages event notifications for connection, disconnection, errors, and new messages.

Delegates

[DASMessageListener.DasErrorHandler](#)

Delegate when the DAS is facing to an error

[DASMessageListener.DasEventHandler](#)

Delegate used for the event related to the DAS

Class DASMessageListener

Namespace: [Teradyne.Archimedes.DAS.Listener](#)

Assembly: Teradyne.Archimedes.DAS.Listener.dll

DASMessageListener is responsible for starting and stopping the DAS HTTP listener, receiving incoming HTTP requests, and dispatching them to a processor. It manages event notifications for connection, disconnection, errors, and new messages.

```
public class DASMessageListener : IDisposable
```

Inheritance

[object](#)  ← DASMessageListener

Implements

[IDisposable](#) 

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

DASMessageListener(string)

Initializes a new instance of the DASMessageListener class with a DAS URL.

```
public DASMessageListener(string dasurl)
```

Parameters

dasurl [string](#) 

The URL to listen to.

Exceptions

[ArgumentNullException](#) 

Fields

_messageList

System for managing the list of messages we want to receive

```
protected AMPMessageManager _messageList
```

Field Value

[AMPMessageManager](#)

Properties

DASURL

Gets the DAS listener URL.

```
public string DASURL { get; }
```

Property Value

[string](#) 

Methods

Dispose()

Disposes the listener and stops it if necessary.

```
public void Dispose()
```

Start()

Starts the HTTP listener and begins processing incoming requests.

```
public Task<bool> Start()
```

Returns

[Task](#) [<bool>](#)

True if started successfully, false otherwise.

Stop()

Stops the HTTP listener and cancels all ongoing tasks.

```
public void Stop()
```

SubscribeToMessages(List<string>, bool)

Subscribes to a set of message names, optionally resetting previous subscriptions.

```
public void SubscribeToMessages(List<string> messageNames, bool resetFirst = true)
```

Parameters

messageNames [List](#) [<string>](#)

List of message names to subscribe to.

resetFirst [bool](#)

Whether to clear existing subscriptions first.

Events

Connected

Event when the DAS is connected

```
public event DASMessageListener.DasEventHandler Connected
```

Event Type

[DASMessageListener.DasEventHandler](#)

Disconnected

Event when the DAS is disconnected

```
public event DASMessageListener.DasEventHandler Disconnected
```

Event Type

[DASMessageListener.DasEventHandler](#)

Error

Event when the DAS encountered an error

```
public event DASMessageListener.DasErrorHandler Error
```

Event Type

[DASMessageListener.DasErrorHandler](#)

NewMessage

Event when the DAS received a new message

```
public event AMPEvents.NewMessageDlg NewMessage
```

Event Type

[AMPEvents.NewMessageDlg](#)

Delegate DASMessageListener.DasErrorHandler

Namespace: [Teradyne.Archimedes.DAS.Listener](#)

Assembly: Teradyne.Archimedes.DAS.Listener.dll

Delegate when the DAS is facing to an error

```
public delegate void DASMessageListener.DasErrorHandler(Exception exception)
```

Parameters

exception [Exception](#) 

Delegate

DASMessageListener.DasEventHandler

Namespace: [Teradyne.Archimedes.DAS.Listener](#)

Assembly: Teradyne.Archimedes.DAS.Listener.dll

Delegate used for the event related to the DAS

```
public delegate void DASMessageListener.DasEventHandler()
```



 Description	 Download Link
Entire Solution	The solution
GIT Hub Link	Git Hub 

More details about those source [code here](#)

This documentation is part of the Teradyne Archimedes Reference Architecture series.